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This project focused on predicting the stock price of **NVIDIA Corporation (NVDA)** by combining two types of data: historical stock prices and **public sentiment from Twitter**. The goal was to see if people's opinions shared online could help make stock predictions more accurate.

To understand sentiment, I used a pre-trained language model called **RoBERTa**, which analyzed tweets and classified them as positive, negative, or neutral. These sentiment scores were then combined with daily stock market data.

For the prediction part, I used a **Long Short-Term Memory (LSTM)** model—an advanced type of neural network that works well with time-based data. I built two versions:

1. One that used only stock prices.
2. One that also included the Twitter sentiment.

Both models performed similarly in terms of accuracy (measured by Mean Squared Error). However, some interesting patterns were found:

- There was a **strong link between tweet volume and stock trading volume** (correlation of 0.72), meaning when more people tweeted about NVIDIA, more people also traded its stock.
- A few specific cases showed that **positive sentiment on Twitter matched sharp increases in stock price**, suggesting that sentiment may help explain short-term changes.

Even though the overall prediction accuracy didn't improve much with sentiment, the analysis showed that **social media can give useful signals during active or volatile market periods**. This approach could be helpful if combined with other market indicators or more advanced models in the future.